Performance bounds for coupled CP model in the framework of hyperspectral super-resolution

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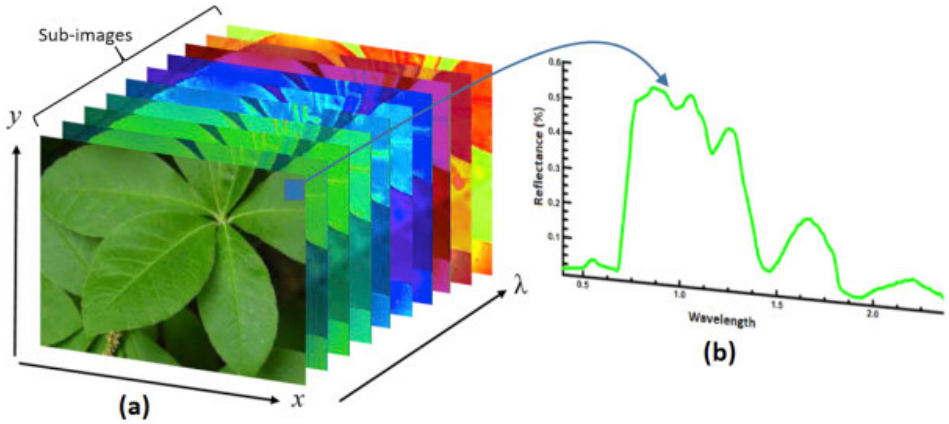
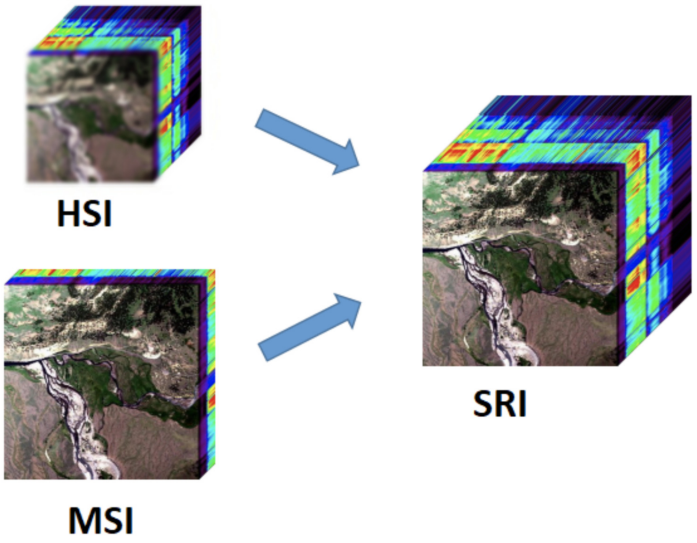
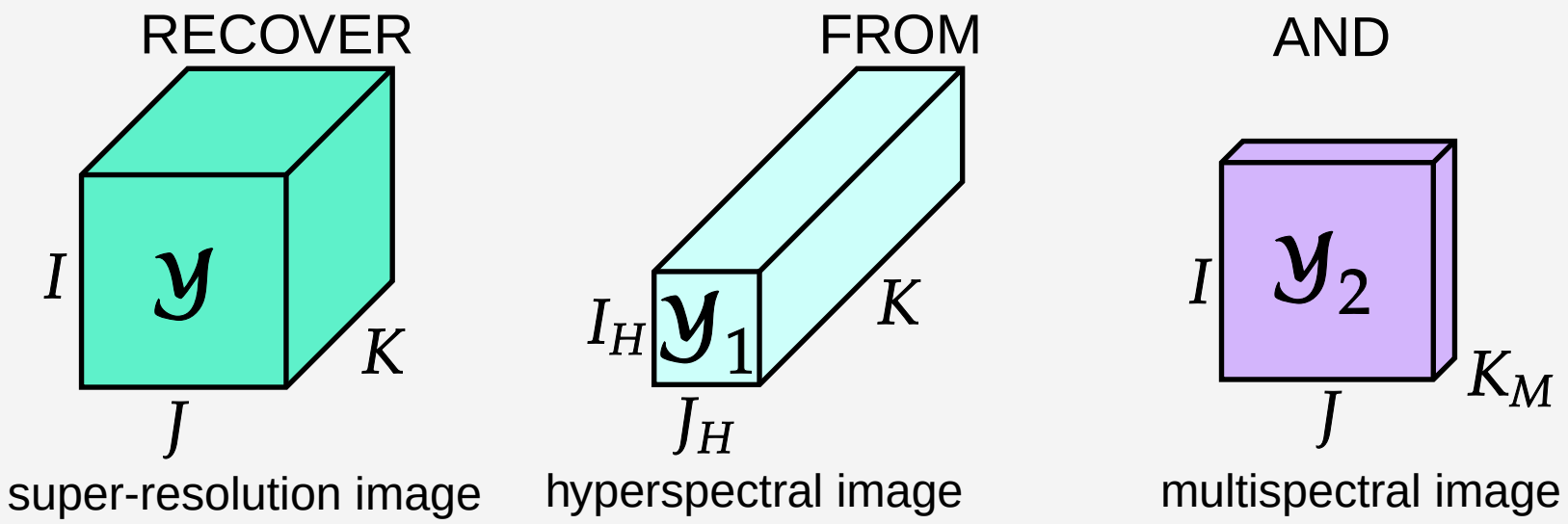
Hyperspectral images





Context

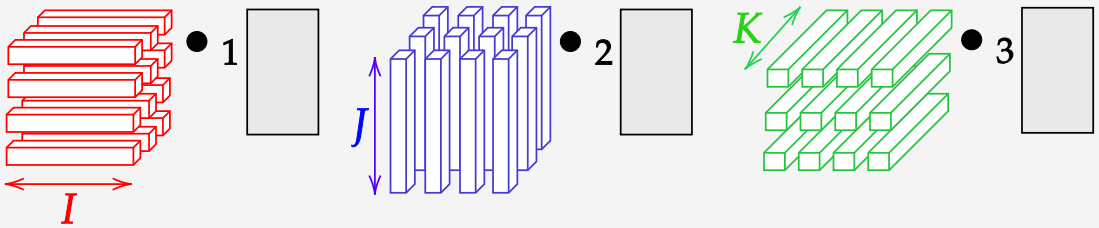
Hyperspectral super-resolution (HSR)



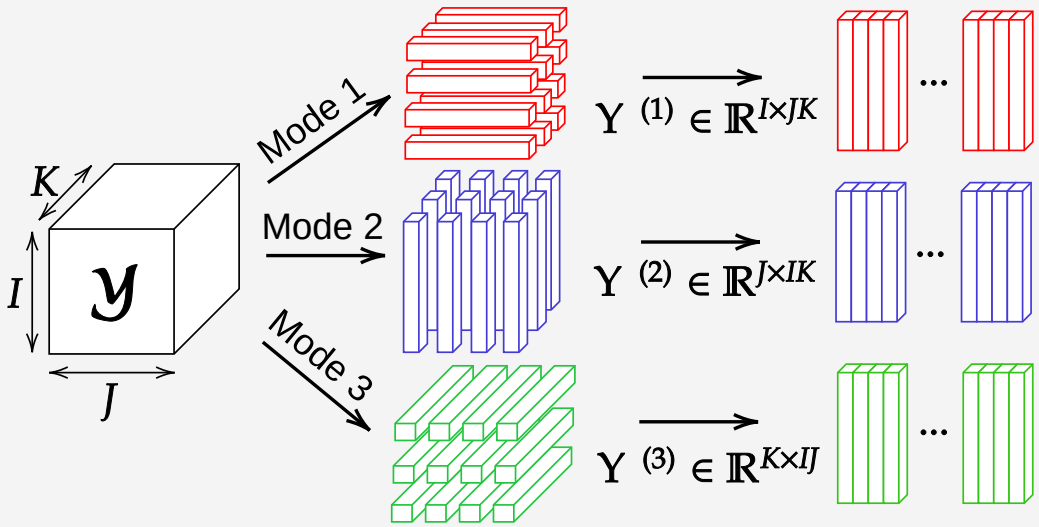
- $\mathcal{Y}_1$: low *spatial* resolution $(I_H < I,\ J_H < J)$;
- $\mathcal{Y}_2$: low *spectral* resolution $(K_M \ll K)$.

Basic tensor operations

Mode product

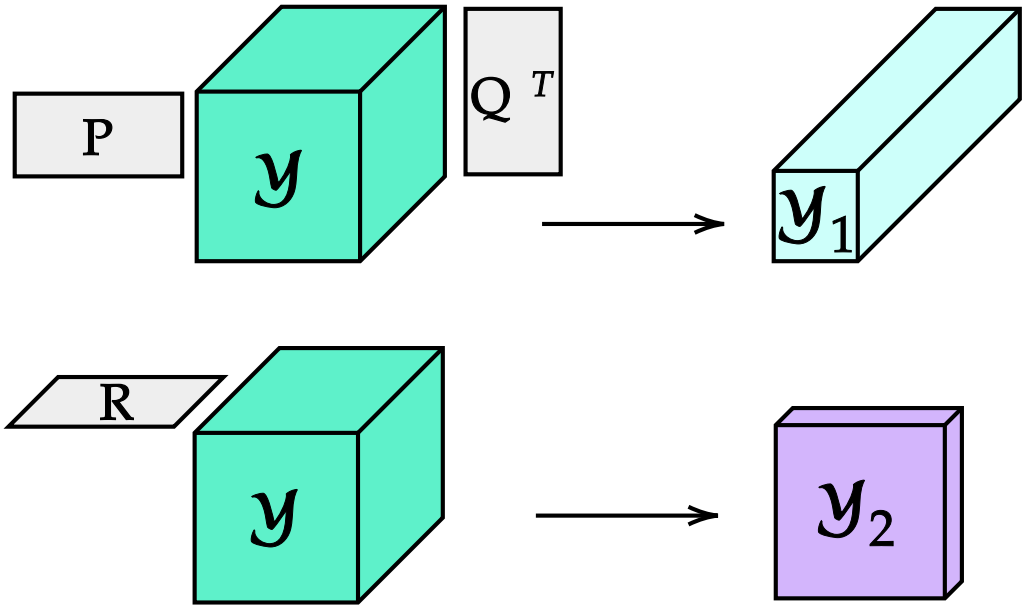


Unfoldings



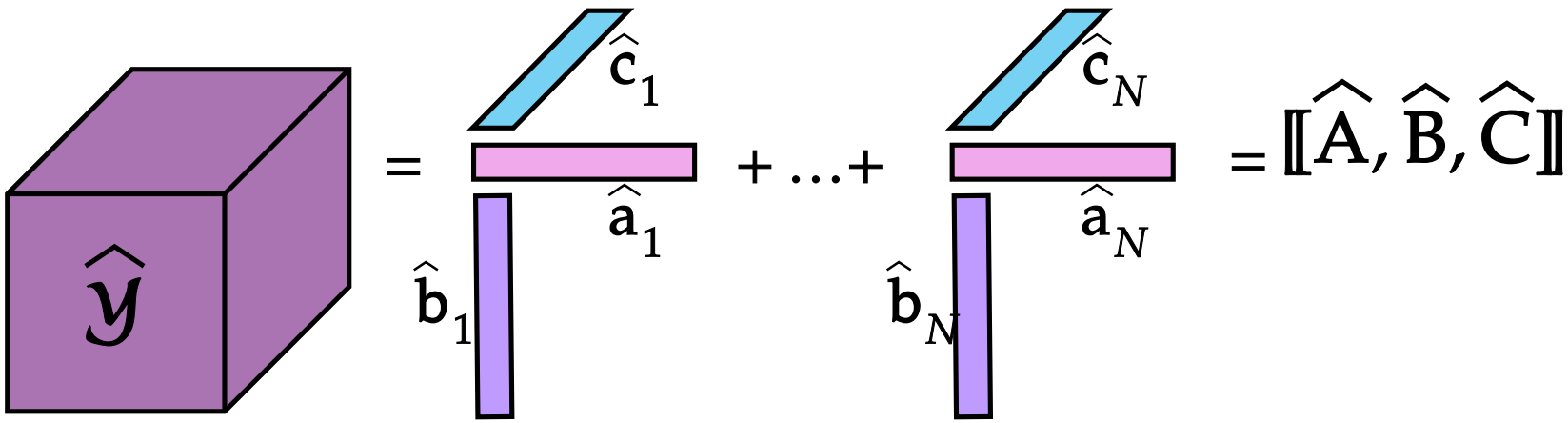
Degradation model

$$\begin{cases} \mathcal{Y}_1 &= \mathcal{Y} \bullet_1 \boldsymbol{P} \bullet_2 \boldsymbol{Q} + \mathcal{E}_1, \\ \mathcal{Y}_2 &= \mathcal{Y} \bullet_3 \boldsymbol{R} + \mathcal{E}_2. \end{cases}$$



- $\boldsymbol{P}, \boldsymbol{Q}, \boldsymbol{R}$: fat degradation matrices ;
- $\mathcal{E}_1 \sim \mathcal{N}(0, \sigma_1^2 \boldsymbol{I})$, $\mathcal{E}_2 \sim \mathcal{N}(0, \sigma_2^2 \boldsymbol{I})$.

Coupled CP model



$$\begin{cases} \mathcal{Y}_1 &= [\![\boldsymbol{A}_1, \boldsymbol{B}_1, \boldsymbol{C}_1]\!] + \boldsymbol{\mathcal{E}}_1, \\ \mathcal{Y}_2 &= [\![\boldsymbol{A}_2, \boldsymbol{B}_2, \boldsymbol{C}_2]\!] + \boldsymbol{\mathcal{E}}_2, \end{cases}$$

subject to $\boldsymbol{A}_1 = \boldsymbol{P}\boldsymbol{A}_2, \boldsymbol{B}_1 = \boldsymbol{Q}\boldsymbol{B}_2, \boldsymbol{C}_2 = \boldsymbol{R}\boldsymbol{C}_1$.

- STEREO : coupled ALS algorithm ;
- $\mathcal{Y} = [\![\boldsymbol{A}_2, \boldsymbol{B}_2, \boldsymbol{C}_1]\!]$.

Estimation

Criterion to minimize

$$\|\boldsymbol{\mathcal{Y}}_1 - [\![\boldsymbol{A}_1, \boldsymbol{B}_1, \boldsymbol{C}_1]\!]\|_F^2 + \lambda \|\boldsymbol{\mathcal{Y}}_2 - [\![\boldsymbol{A}_2, \boldsymbol{B}_2, \boldsymbol{C}_2]\!]\|_F^2,$$
$$\text{subject to } \boldsymbol{A}_1 = \boldsymbol{P}\boldsymbol{A}_2, \boldsymbol{B}_1 = \boldsymbol{Q}\boldsymbol{B}_2, \boldsymbol{C}_2 = \boldsymbol{R}\boldsymbol{C}_1.$$

- $\lambda = \frac{\sigma_1^2}{\sigma_2^2} \rightarrow$ Log-likelihood criterion ;
  $\rightarrow$ minimizer = MLE.

Model parameters [Ren et al., 2015]

$$\boldsymbol{\theta}_1 = \mathrm{vec}\{\boldsymbol{C}_1\} \in \mathbb{R}^{KN\times 1}, \qquad \boldsymbol{\phi}_1 = \begin{bmatrix}\mathrm{vec}\{\boldsymbol{A}_1\}\\ \mathrm{vec}\{\boldsymbol{B}_1\}\end{bmatrix} \in \mathbb{R}^{(I_H+J_H)N\times 1},$$

$$\boldsymbol{\theta}_2 = \mathrm{vec}\{\boldsymbol{C}_2\} \in \mathbb{R}^{K_M N\times 1}, \quad \boldsymbol{\phi}_2 = \begin{bmatrix}\mathrm{vec}\{\boldsymbol{A}_2\}\\ \mathrm{vec}\{\boldsymbol{B}_2\}\end{bmatrix} \in \mathbb{R}^{(I+J)N\times 1}.$$

- HSI : only degraded *spatially*;
- MSI : only degraded *spectrally*;

General coupled model

General coupled model

$$\begin{cases} \boldsymbol{\mathcal{Y}}_1 \sim f_{\boldsymbol{\mathcal{Y}}_1;\boldsymbol{\theta}_1,\boldsymbol{\phi}_1} \text{ and } \boldsymbol{\mathcal{Y}}_2 \sim f_{\boldsymbol{\mathcal{Y}}_2;\boldsymbol{\theta}_2,\boldsymbol{\phi}_2} \\ g(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) = 0 \text{ and } h(\boldsymbol{\phi}_1, \boldsymbol{\phi}_2) = 0. \end{cases}$$

Aim : performance analysis for $\boldsymbol{\theta}_1$, $\boldsymbol{\phi}_1$, $\boldsymbol{\theta}_2$, $\boldsymbol{\phi}_2$.

- MLE for constrained parameters : achieves the CCRB [Gorman and Hero, 1990]

Coupled CP model

Specific case : $\boldsymbol{P}$, $\boldsymbol{Q}$, $\boldsymbol{R}$ are subsampling matrices ;

$$\boldsymbol{P} = \begin{bmatrix} \mathbf{1} & 0 & 0 & \dots & \dots & 0 \\ 0 & 1 & 0 & \dots & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & & \vdots \\ 0 & \dots & 0 & 1 & \dots & 0 \end{bmatrix}, \quad \boldsymbol{Q} = \begin{bmatrix} \mathbf{1} & 0 & 0 & \dots & \dots & 0 \\ 0 & 0 & 1 & \dots & \dots & 0 \\ \vdots & \ddots & \ddots & & \ddots & \vdots \\ 0 & \dots & 0 & 0 & 1 & 0 \end{bmatrix}$$

$\rightarrow$ **Necessary** : first row $= \begin{bmatrix} \mathbf{1} & 0 & \dots & 0 \end{bmatrix}$.

Identifiability of CP model

Assumption : identifiability of CPDs of $\mathcal{Y}_1$ and $\mathcal{Y}_2$.

$\rightarrow$ CPD : unique up to permutation and scaling ambiguities.

- Permutation (*uncoupled model*) : best permutation of $\boldsymbol{C}_2$, fixed $\boldsymbol{C}_1$;
- Scaling : $\boldsymbol{A}_i(:,1) = \boldsymbol{B}_i(:,1) = 1$;
  $\rightarrow$ valid for such **subsampling** $\boldsymbol{P}$, $\boldsymbol{Q}$, $\boldsymbol{R}$.

$$\widetilde{\boldsymbol{\phi}}_1 = \boldsymbol{M}_1\boldsymbol{\phi}_1 \in \mathbb{R}^{(I_H+J_H-2)N} \qquad \widetilde{\boldsymbol{\phi}}_2 = \boldsymbol{M}_2\boldsymbol{\phi}_2 \in \mathbb{R}^{(I+J-2)N}$$

- $\boldsymbol{M}_i$: selection matrices.

Uncoupled model

Standard CRB

$$\begin{cases} \mathrm{vec}\{\boldsymbol{\mathcal{Y}}_1\} \sim \boldsymbol{\mathcal{N}}(\mu_1(\boldsymbol{\theta}_1, \widetilde{\boldsymbol{\phi}}_1), \sigma_1^2 \boldsymbol{I}), \\ \mathrm{vec}\{\boldsymbol{\mathcal{Y}}_2\} \sim \boldsymbol{\mathcal{N}}(\mu_2(\boldsymbol{\theta}_2, \widetilde{\boldsymbol{\phi}}_2), \sigma_2^2 \boldsymbol{I}), \end{cases}$$

$$\mu_1(\boldsymbol{\theta}_1, \widetilde{\boldsymbol{\phi}}_1) \;=\; \mathrm{vec}\{[\![\boldsymbol{A}_1, \boldsymbol{B}_1, \boldsymbol{C}_1]\!]\},\; \mu_2(\boldsymbol{\theta}_2, \widetilde{\boldsymbol{\phi}}_2) \;=\; \mathrm{vec}\{[\![\boldsymbol{A}_2, \boldsymbol{B}_2, \boldsymbol{C}_2]\!]\}.$$

- $\boldsymbol{F}(\boldsymbol{\theta}_i, \widetilde{\boldsymbol{\phi}}_i)$ $(i = 1, 2)$: Slepian-Bangs formula [Slepian., 1954] ;
- $\mathbf{CRB}(\boldsymbol{\theta}_i, \widetilde{\boldsymbol{\phi}}_i) = \boldsymbol{F}^{-1}(\boldsymbol{\theta}_i, \widetilde{\boldsymbol{\phi}}_i)$;
- Full derivation in [Ren et al., 2016].

Coupled model

Constrained CRB (CCRB)

$$\mathbf{CCRB} = \boldsymbol{F}^{-1} - \boldsymbol{F}^{-1}\boldsymbol{J}^{\mathsf{T}}\left(\boldsymbol{J}\boldsymbol{F}^{-1}\boldsymbol{J}^{\mathsf{T}}\right)^{-1}\boldsymbol{J}\boldsymbol{F}^{-1} \succeq \mathbf{0},$$

$$\boldsymbol{F} \triangleq \mathrm{Diag}\{\boldsymbol{F}(\boldsymbol{\theta}_1, \widetilde{\boldsymbol{\phi}}_1), \boldsymbol{F}(\boldsymbol{\theta}_2, \widetilde{\boldsymbol{\phi}}_2)\}, \qquad \boldsymbol{J} \triangleq \boldsymbol{J}(\boldsymbol{\theta}_1, \widetilde{\boldsymbol{\phi}}_1, \boldsymbol{\theta}_2, \widetilde{\boldsymbol{\phi}}_2).$$

- $\boldsymbol{J}$: Jacobian of the constraints.

$$\boldsymbol{J} = \begin{bmatrix} \frac{\partial g(\boldsymbol{\theta}_1,\boldsymbol{\theta}_2)}{\partial \boldsymbol{\theta}_1^{\mathsf{T}}} & \mathbf{0} & \frac{\partial g(\boldsymbol{\theta}_1,\boldsymbol{\theta}_2)}{\partial \boldsymbol{\theta}_2^{\mathsf{T}}} & \mathbf{0} \\ \mathbf{0} & \frac{\partial h(\widetilde{\boldsymbol{\phi}}_1,\widetilde{\boldsymbol{\phi}}_2)}{\partial \widetilde{\boldsymbol{\phi}}_1^{\mathsf{T}}} & \mathbf{0} & \frac{\partial h(\widetilde{\boldsymbol{\phi}}_1,\widetilde{\boldsymbol{\phi}}_2)}{\partial \widetilde{\boldsymbol{\phi}}_2^{\mathsf{T}}} \end{bmatrix}.$$

Constraints formulation

$$\begin{cases} g(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) & = (\boldsymbol{I}_N \boxtimes \boldsymbol{R})\, \boldsymbol{\theta}_1 - \boldsymbol{I}_{K_M N} \boldsymbol{\theta}_2 = 0, \\ h(\widetilde{\boldsymbol{\phi}}_1, \widetilde{\boldsymbol{\phi}}_2) & = \boldsymbol{I}_{(I_H + J_H - 2)N} \widetilde{\boldsymbol{\phi}}_1 \\ & \quad - \boldsymbol{M}_1 \underbrace{\begin{bmatrix} \boldsymbol{I}_N \boxtimes \boldsymbol{P} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I}_N \boxtimes \boldsymbol{Q} \end{bmatrix}}_{\boldsymbol{\mathcal{P}}} \boldsymbol{M}_2^{\mathsf{T}} \widetilde{\boldsymbol{\phi}}_2 = 0, \end{cases}$$

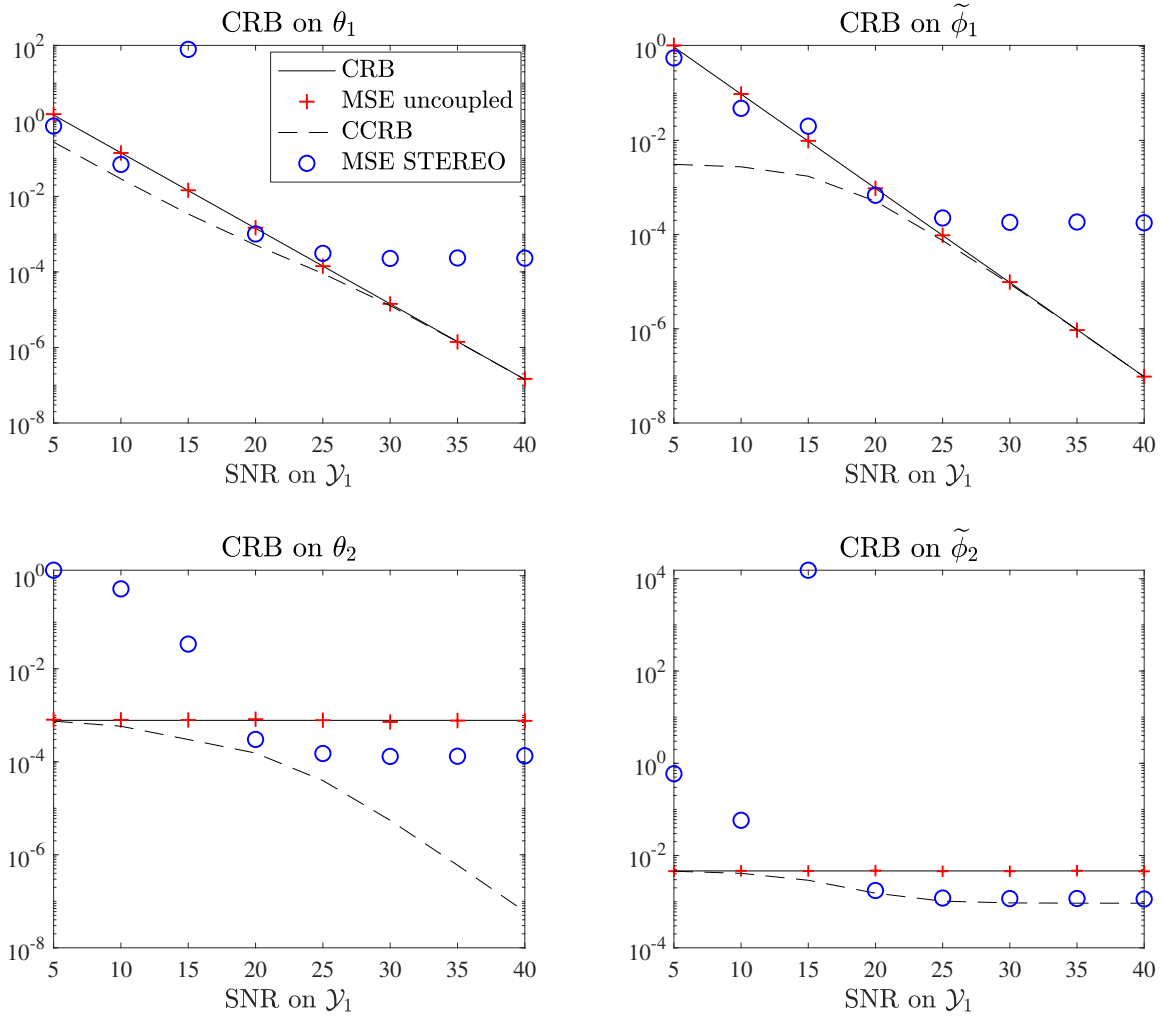
$$\boldsymbol{J} = \begin{bmatrix} \boldsymbol{I}_N \boxtimes \boldsymbol{R} & \boldsymbol{0} & -\boldsymbol{I}_{K_M N} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I}_{(I_H + J_H - 2)N} & \boldsymbol{0} & -\boldsymbol{M}_1 \boldsymbol{\mathcal{P}} \boldsymbol{M}_2^{\mathsf{T}} \end{bmatrix}.$$

Simulations setup

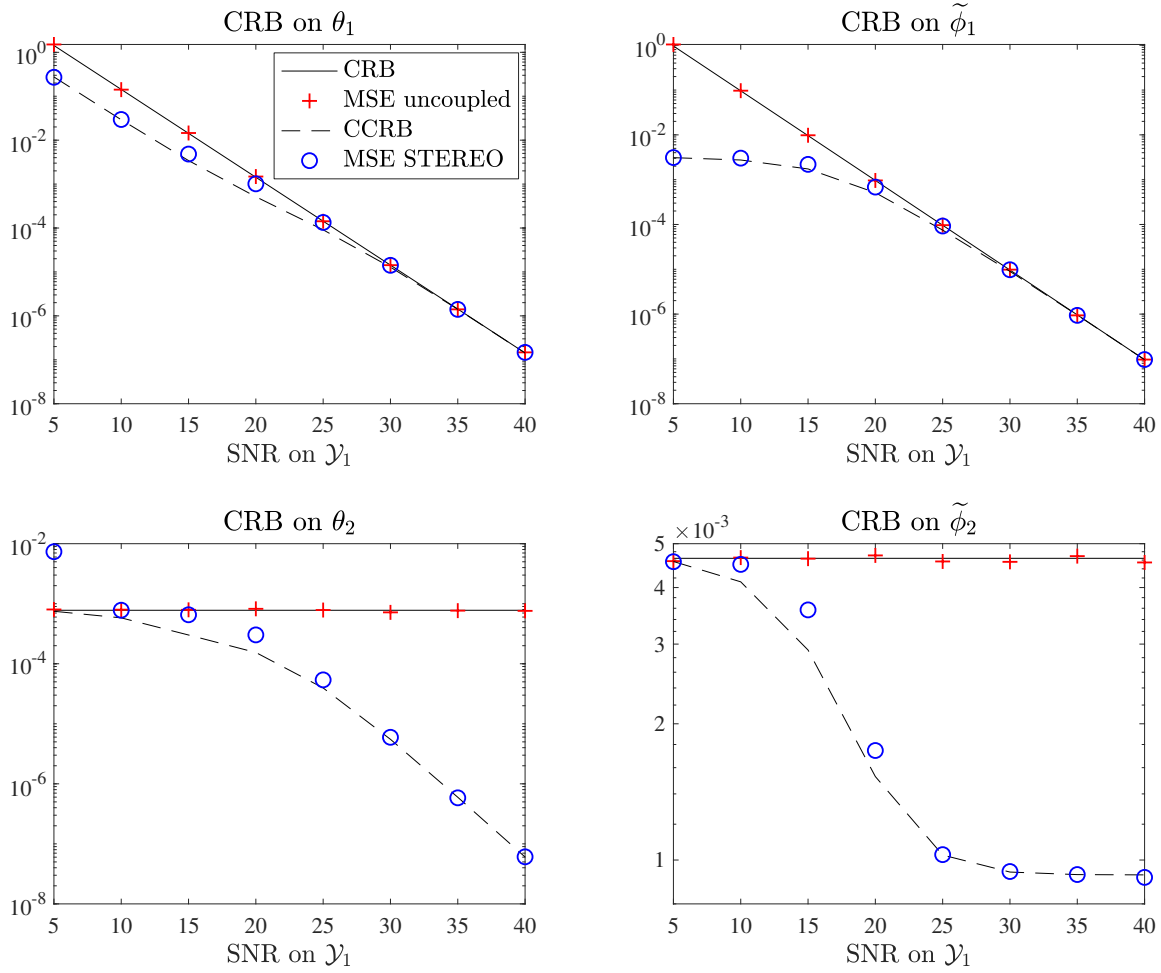
What is the influence of the parameter λ on the performance of STEREO ?

- $I = J = K = 15$, $I_H = J_H = 10$, $K_M = 9$, $N = 3$;
- $\boldsymbol{A}_i, \boldsymbol{B}_i, \boldsymbol{C}_i$: i.i.d. real Gaussian variables ;
- $\boldsymbol{P}, \boldsymbol{Q}, \boldsymbol{R}$: generated from identity matrices ;
- ALS + STEREO (500 iter. max.) : MSE over 500 realizations ;
- SNR on $\boldsymbol{\mathcal{Y}}_1$: 5 to 40dB ; SNR on $\boldsymbol{\mathcal{Y}}_2$: 20dB.

Results with $\lambda = 1$



Results with $\lambda = \frac{\sigma_1^2}{\sigma_2^2}$



Conclusion

- Choice of λ : data-driven procedures [Hansen, 1999] or robust methods [Roemer et al., 2013] ;
- Total MSE for STEREO : almost reaches the CCRB for $\lambda = \frac{\sigma_1^2}{\sigma_2^2}$;

$$\|\boldsymbol{\mathcal{Y}}_1 - [\![\boldsymbol{A}_1, \boldsymbol{B}_1, \boldsymbol{C}_1]\!]\|_F^2 + \lambda \|\boldsymbol{\mathcal{Y}}_2 - [\![\boldsymbol{A}_2, \boldsymbol{B}_2, \boldsymbol{C}_2]\!]\|_F^2,$$

s. to $\boldsymbol{A}_1 = \boldsymbol{P}\boldsymbol{A}_2, \boldsymbol{B}_1 = \boldsymbol{Q}\boldsymbol{B}_2, \boldsymbol{C}_2 = \boldsymbol{R}\boldsymbol{C}_1$

$$\neq$$

$$\underbrace{\|\boldsymbol{\mathcal{Y}}_1 - [\![\boldsymbol{P}\boldsymbol{A}_2, \boldsymbol{Q}\boldsymbol{B}_2, \boldsymbol{C}_1]\!]\|_F^2 + \lambda \|\boldsymbol{\mathcal{Y}}_2 - [\![\boldsymbol{A}_2, \boldsymbol{B}_2, \boldsymbol{R}\boldsymbol{C}_1]\!]\|_F^2}_{\text{criterion minimized by STEREO}}.$$

Perspectives

- Derivation of performance bounds **adapted to STEREO** ;
- Extension to **more realistic degradation matrices** (blurring and downsampling, averaging).

Thank you for your attention !